

TRANSPORT LAYER

The transport layer is the fourth layer of the OSI reference model. Transport layer provides transparent, reliable, and cost effective transfer of data units between the upper layer entities in the end systems.

The goal of the transport layer is to abstract the structure and function of the network so that the upper layer can communicate without needing to consider the network technology.

Functions of Transport Layer

1: End-to-End Delivery: The network layer oversees the end-to-end delivery of individual packets but does not see any relationship between those packets, even those belonging to a single message. It treats each as an independent entity

2: Addressing: The client needs the address of the remote computer it wants to communicate with. Such remote computers have a unique address so that it can be distinguished from all the other computers.

3: Reliable Delivery: The reliable delivery considers the following issues given below:

*Error Control: When transferring data, the primary goal of reliability is error control. Data must be delivered to their destination exactly as they originated from the source. The realities of physical data transport are that, while 100 percent error-free delivery is probably impossible, transport layer protocols are designed to come as close as possible.

*Sequence Control: The second aspect of reliability implemented at the transport layer is sequence control. On the sending end, the transport layer is responsible for ensuring that data units received from the upper layers are usable by the lower layers.

On the receiving end, it is responsible for ensuring that the various pieces of a transmission are correctly reassembled.

*Loss Control: The third aspect of reliability covered by the transport layer is loss control. The transport layer ensures that all pieces of a transmission arrive at the destination, not just some of them.

*Duplication Control: The fourth aspect of reliability covered by the transport layer is duplication control. Transport layer functions must guarantee that no pieces of data arrive at the receiving system duplicated.

4: Flow Control: Fast host cannot keep pace with a slow one. Hence, this is a mechanism to regulate the flow of information. The amount of memory on a computer is limited, and without flow control a larger computer might flood a computer with so much information that it can't hold it all before dealing with it. Nowadays, this is not a big issue, as memory is cheap while bandwidth is comparatively expensive, but in earlier times it was more important. Flow control allows the receiver to respond before it is overwhelmed

5: Multiplexing: To improve transmission efficiency, the transport layer has the option of multiplexing.

TRANSMISSION CONTROL PROTOCOL (TCP)

TCP is a connection-oriented reliable protocol. It provides a reliable transport service between pairs of processes executing on End Systems (ES) using the network layer service provided by the IP protocol it is the general protocol suite of the Internet; encompassing protocols for network activities such as datagram delivery and acknowledgement and protocols for user activity such as remote login (Telnet) and file transfer protocol (FTP). A given Implementation of TCP/IP application typically contain a dozen or more protocols and facilities essential to the use of TCP/IP application in a TCP/IP environment. It is a 2-layer predominant transfer protocol that provides a reliable connection oriented byte stream service

TCP PACKET FORMAT

The following description summarise the TCP packet fields illustrated in below

1: Source Port and Destination Port: Identifies points at which upper-layer source and destination processes receive TCP service

2: Sequence Number: Usually specifies the number assigned to the first byte of data in the current message. In the connection-establishment phase, this field also can be used to identify an initial sequence number to be used in an upcoming transmission.

3: Acknowledgment Number: Contains the sequence number of the next byte of data the sender of the packet expects to receive

4: Data Offset: Indicates the number of 32-bit words in the TCP header.

5: Reserved: Remains reserved for future use.

6: Flags(6 bits): For each flag, if set to 1, the meaning is as follows:

- *CWR: Congestion window reduced.
- *ECE: ECN-Echo: the CWR and ECE bits, defined in RFC 3168 are used for the explicit congestion notification function.
- *URG: Urgent pointer field significant.
- *ACK: Acknowledgment field significant.
- *PSH: Push function.
- *RST: Reset the connection.
- *SYN: Synchronise the sequence number
- *FIN: No more data from sender.

7: Window: Specifies the size of the sender's receive window (that is the buffer space available for incoming data).

8: Checksum: indicates whether the header was damaged in transit

9: Urgent Pointer: Points to the first urgent data byte in the packet.

10: Options: Specifies various TCP options.

11: Data: Contains upper-layer information

Source Port			Destination Port		
Sequence Number					
Acknowledgement Number					
Data Offset	Reserved	Flags		Window	
Checksum			Urgent Pointer		
Options (+padding)					
Data (variable)					

TCP SERVICE

There is a long list of services that can be optionally provided by the Transport Layer. All available services are

1: Connection-Oriented: This is normally easier to deal with than connection-less models. so where the Network layer only provide a connection-less service, often a connection-oriented service is built on top of that in the Transport Layer

2: Reliable Data: Packets may be lost in routers, switches, bridges and hosts due to network congestion. when the packet queues are filled and the network nodes have to delete packets. Packets may be lost or corrupted in Ethernet due to interference and noise, since Ethernet does not retransmit corrupted packets. Packets may be delivered in the wrong order by an underlying network.

3: flow Control: The amount of memory on a computer is limited. and without flow control a larger computer might flood a computer with so

much information that it can't hold it all before dealing with it. Flow control allows the receiver to respond before it is overwhelmed.

4: Congestion Avoidance: Network congestion occurs when a queue buffer of a network node is full and starts to drop packets. Automatic repeat request may keep the network in a congested state. This situation can be avoided by adding congestion avoidance to the flow control. Including slow-start.

5: Byte Orientation: The Transport Layer may add the ability to view communication just as a stream of bytes. This is nicer to deal with than random packet sizes, however, it rarely matches the communication model which will normally be a sequence of messages of user defined sizes.

6: Ports: Ports are essential ways to address multiple entities in the same location. For example, the first line of a postal address is a kind of port and distinguishes between different occupants of the same house.

UDP: USER DATAGRAM PROTOCOL

UDP is one of the core members of the Internet protocol suite. The protocol was designed by David P.

UDP uses a simple connectionless transmission model with a minimum of protocol mechanism. It has no handshaking dialogues, and thus exposes any unreliability of the underlying network protocol to the user's program. There is no guarantee of delivery, ordering, or duplicate protection. UDP provides checksums for data integrity, and port numbers for addressing different functions at the source and destination of the datagram.

With UDP, computer applications can send messages, in this case referred to as datagram, to other hosts on an Internet Protocol (IP) network without prior communications to set up special transmission channels or data paths. UDP is suitable for purposes where error checking and correction is either not necessary or is performed in the application, avoiding the overhead of such processing at the network interface level.

UDP provides two services not provided by the IP layer. It provides port numbers to help distinguish different user requests and, optionally, a checksum capability to verify that the data arrived intact.

UDP Header Format

A computer may send UDP Packets without first establishing a connection to the recipient. the computer completes the appropriate fields in the UDP header (PCI) and forward the data together with the header for transmission by the IP network layer

The UDP Header consist of four fields each of 2 bytes i length

1: Source Port: UDP Packets from a client use this a service access point (SAP) to indicate the session on the local client that originated the Packet. UPD Packet from a server carry the server SAP in the field.

2: Destination Port: UDP Packets from a client use this a service access point (SAP) to indicate the service required from the remote server. UPD Packet from a server carry the client SAP in the field.

3: UDP Length: The Number of bytes comprising the combined UDP Header information and payload data

4: UDP Checksum: A Checksum to verify that the end to end data has not been corrupted by routers or bridges in the network or by the processing in an end system. The algorithm to compute the Checksum is the standard internet Checksum algorithm. If this check is not required, the value of 0x0000 is placed in the field in which case the data is not checked by the receiver

APPLICATION LAYER

The application layer is the OSI layer closest to the end user. which means that both the OSI application layer and the user interact directly with the software application. This layer in1eracts with software applications that implement a communicating component. Such application programs fall outside the scope of the OSI model.

Functions of Application Layers

1: File Transfer, Access, and Management (FTAM): This application allows a user to access files in a remote computer (to make changes or read data), to retrieve files from a remote computer: and to manage or control files in a remote computer

2: Addressing: For communication between client and server there is requirement of address. When the client request is made to server, it also contains server address and us

own address. The server response also contains destination address i.e, the address of the client. DNS is used for this addressing.

3: Mail Services: This Application provides the basic for email forwarding the storage

4: Directory Services: This Application provides distributed database source and access for global information about various objects and services

5: Authentication: Authenticates the sender or receiver of the message or both

FILE TRANSFER PROTOCOL (FTP)

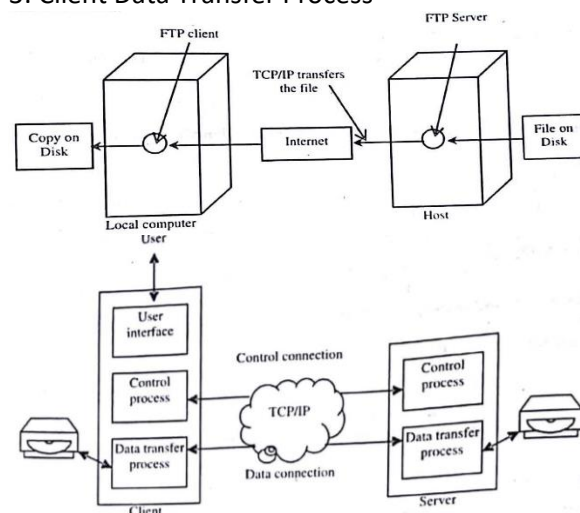
FTP allows the transfer on files one computer to another. file can be in any format like text, graphics, sound etc. it activates the client-server relationship. thus whatever that can be stored in a computer can be moved with the FTP service

Objectives of FTP

- 1: Its main objective is to help in sharing of programs and data
- 2: Inspiring the implicit use of remote computer
- 3: To protect the user from variation in file storage system among numerous hosts
- 4: Effective an reliable sharing of data

Mechanism of FTP

- 1: User Interface, 2: Client Control Process
- 3: Client Data Transfer Process



FTP Has two Connections

- 1: Control Connection: this is a Control process Connection. during the complete FTP session, control Connections remain Connected
- 2: Data Connection: this Connection is present b/w various data transfer processes, during every file transfer this Connection opens and after completion of transfer process, this Connection gets closed

DOMAIN NAME SPACE (DNS)

DNS is a directory lookup service that provides a mapping between the name of a host on the Internet and its numerical address. DNS is essential to the functioning of the Internet. DNS is a service on a TCP/IP network which allows users of networks to utilise user-friendly names when looking for other hosts (computers) instead of having to remember and use their IP Addresses. This system is used extensively on the Internet and in many private enterprises today.

Format of Domain Names

Host Name Or Sub Domain	Domain/Sub Domain	Top-Level Domain
www	amazon	com

Domain Name Servers

In theory at least a single name server could contain the entire DNS database and respond to all queries about it. In practice, this server could be so overloaded as to be useless. Furthermore, if it ever went down, the entire Internet would be crippled. To avoid the problems associated with having only a single source of information, the DNS name space is divided up into non-overlapping zones. Each zone contains some part of the tree and also contains name servers holding the authoritative information about that zone.

ELECTRONIC MAIL

One of the most popular Internet services is electronic mail (e-mail). The designers of the Internet probably never imagined the popularity of this application program. At the beginning of the Internet era, the messages sent by electronic mail were short and consisted of text only; they let people exchange quick memos. Today, electronic mail is much more complex. It allows a message to include text, audio, and video. It also allows one message to be sent to one or more recipients.

Architecture

User Agent: The first component of an electronic mail system is the user agent. It provides service to the user to make the process of sending and receiving a message easier.

Composing Messages

A user agent helps the user compose the e-mail message to be sent out. Most user agents provide a template on the screen to be filled in

by the user. Some even have a built-in editor that can do spell checking, grammar checking, and other tasks expected from a sophisticated word processor. A user, of course, could alternatively use his or her favourite text editor or word processor to create the message and import it, or cut and paste it, into the user agent template.

Reading Messages : The second duty of the user agent is to read the incoming messages. When a user invokes a user agent, it first checks the mail in the incoming mailbox. Most user agents show a one-line summary of each received mail. Each e-mail contains the following fields.

1. A number field.
2. A flag field that shows the status of the mail such as new, already read but not replied to, or read and replied to.
3. The size of the message.
4. The sender.
5. The optional subject field.

Replying to Messages After reading a message, a user can use the user agent to reply to a message. A user agent usually allows the user to reply to the original sender or to reply to all recipients of the message. The reply message may contain the original message and the new message.

Forwarding Messages Replying is defined as sending a message to the sender or recipients of the copy. Forwarding is defined as sending the message to a third party. A user agent allows the receiver to forward the message, with or without extra comments, to a third party.

Handling Mailboxes A user agent normally creates two mailboxes: an inbox and an outbox. Each box is a file with a special format that can be handled by the user agent. The inbox keeps all the received e-mails until they are deleted by the user. The outbox keeps all the sent e-mails until the user deletes them. Most user agents today are capable of creating customized mailboxes.

User Agent Types

Command-Driven: Command-driven user agents belong to the early days of electronic mail. They are still present as the underlying user agents in servers. A command-driven user agent normally accepts a one-character command from the keyboard to perform its task.

MULTIPURPOSE INTERNET MAIL

EXTENSIONS (MIME) is an Internet standard that extends the format of email to support:

1. Text in character sets other than ASCII
2. Non-text attachments: audio, video, images, application programs etc.
3. Message bodies with multiple parts
4. Header information in non-ASCII character sets

Virtually all human-written Internet email and a fairly large proportion of automated email is transmitted via SMTP in MIME format.

MIME-Version

The presence of this header indicates the message is MIME-formatted.

Content-Type

This header indicates the Internet media type of the message content, consisting of a type and subtype

MIME, a specification for formatting non-ASCII messages so that they can be sent over the Internet. Many e-mail clients now support MIME, which enables them to send and receive graphics, audio, and video files via the Internet mail system. In addition, MIME supports messages in character sets other than ASCII. MIME (Multi-Purpose Internet Mail Extensions) is an extension of the original Internet e-mail protocol that lets people use the protocol to exchange different kinds of data files on the Internet: audio, video, images, application programs, and other kinds, as well as the ASCII text handled in the original protocol, the Simple Mail Transport Protocol (SMTP).

There are many predefined MIME types, such as GIF graphics files and PostScript files. It is also possible to define your own MIME types. In addition to e-mail applications, Web browsers also support various MIME types. This enables the browser to display or output files that are not in HTML format.

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SNMP

SNMP is a framework for managing devices in an internet using TCP/IP suite. It provides fundamental operations for monitoring and maintaining an internet.

➤ SNMP uses the concept of manager and agent. Manager usually a host controls and monitors a set of agents, usually routers. A management station, called a manager, is a host that runs the SNMP client program. A managed station, called an agent, is a router or host that runs the SNMP server program.

➤ Management is achieved through simple interaction between a manager and an agent. The agent keeps performance information in a database. The manager has access to the values in the database.

Management with SNMP is based on three basic ideas

1. A manager checks an agent by requesting information that reflects the behavior of the agent.
2. A manager forces an agent to perform a task by resetting values in the agent database.
3. An agent contributes the management process by warning the manager of an unusual situation.

Role of SNMP

SNMP has some very specific roles in network management. It defines the format of the packet to be sent from a manager to an agent and vice versa. It also interprets the result and creates statistics. The packet exchange contains the object names (variables) and their status (values). SNMP is response for reading and changing these values.

SNMP messages:

SNMP does not send only a PDU, it embeds the PDU in a message. A message in SNMPv3 is made of four elements: version, header, security parameter and data.

The version, defines the current version (3)

The header contains values for message identification, maximum message size, message flag and a message security model.

The message security parameter is used to create a message digest.

WORLD WIDE WEB(WWW)

WWW is the most popular internet tool. It is also known as Web. With the help of WWW, user can easily access and present the documents on the internet. These documents are stored on the server. In other words, web is a collection of interlinked hypertext documents which are accessed through internet. Hypertext documents are also known as 'Web Pages'. In a set of web pages (websites), the first one is called Home Page. The information is available on WWW via HTTP (Hypertext Transfer Protocol).

A system of 'interlinked hypertext documents' which can be accessed via the internet is referred to as the web. TCP/IP is the fundamental transport mechanism used in the World Wide Web. Various companies set up websites which are collection of web pages. Web uses browsers like Internet Explorer or Google Chrome for accessing the web pages which are text files, created using HTML code. Web pages may also contain JavaScript code and other commands. Each web page may be a union of text, pictures, audio clips, animations and other data which can be electronically presented. Web pages use hyperlinks for navigation on the web.

WWW Architecture

The general architecture of the WWW is 2 tiered. This architecture has a web client which displays the information content and a web server which provides information to the client (when requested).

For e-commerce applications this architecture works as basic. It develops software where functions are spread over the applications server (where applications exist) data servers (where data are stored) and collection of networked PCs (which are used by the information users).

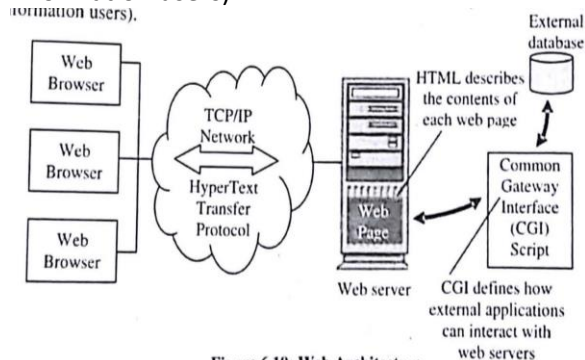


Figure 6.10: Web Architecture

Components of World Wide Web

1: Web Clients: Web clients are also referred to as browsers which provide a graphical user interface for content display. These may run on IBM-compatible PCs, Apple Macintosh computers, UNIX platforms and other platforms which provide a number of facilities. Following are the few most popular browsers:

- *Microsoft's Internet Explorer
- *Firefox Mozilla
- *Google's Chrome.

2: Web Servers: Web servers are group of both software and hardware that act as information storehouses. These programs can be executed over IBM PCs, Apple Macintosh computers, UNIX platforms, etc. The most popular Web clients are Microsoft's Internet Information Server and Netscape's Communications Server.

3: Hypertext Mark-up Language (HTML): This language is used for creating the documents and webpages. It uses a variety of tags and attributes which define the structure and layout of the web document.

4: Uniform Resource Locator (URL): The Uniform Resource Locator is a unique address assigned to a specific Website or web page. This is entered in the address bar of the browser to visit a particular site or page.

5: Hypertext Transfer Language (HTTP): HTTP is the protocol which guides the working of the web. This protocol decides how a message is formatted and transmitted over the internet. It also takes the decision that what will be the action of servers and browsers in response to the different commands.

6: CGI (Common Gateway Interface): CGI is a standard for web server to transfer the web user's request to an application program and also transferring the received data back to the user. For example, when the user requests an URL, the server will immediately display the requested page. But when a user fills out a form and submits it on a web page, it is processed by an application program.